



The problem statement

- Question: Can we design continuous-time Kalman filters?
- **GOAL:** Develop an estimator $\hat{x}(t)$ that is a linear function of measurements $z(\tau)$, $0 \leq \tau \leq t$, and minimizes the function:

$$\mathbb{E}[(x(t) - \hat{x}(t))^T (x(t) - \hat{x}(t))].$$

- **ASSUMED:** Dynamics of linear system are (ignore $u(t)$ for now):

$$\dot{x}(t) = Ax(t) + B_w w(t)$$

$$z(t) = Cx(t) + v(t),$$

where noises are uncorrelated and white and $\hat{x}(0)$ and $\Sigma_{\tilde{x}}(0)$ are given.

- Specifically, $\mathbb{E}[w(t)] = \mathbb{E}[v(t)] = 0$, $\mathbb{E}[w(t_1)w(t_2)^T] = S_w \delta(t_1 - t_2)$,
 $\mathbb{E}[v(t_1)v(t_2)^T] = S_v \delta(t_1 - t_2)$, $\mathbb{E}[w(t_1)v(t_2)] = 0$, $S_w > 0$, $S_v > 0$,
- **APPROACH:** As with some analysis in weeks 2 and 3 of Course #1, analyze discrete-time case as $\Delta t \rightarrow 0$.



Conversion: The Kalman gain $L(t)$

- To convert the Kalman gain, we start with:

$$L_k = \Sigma_{\tilde{x},k}^- C^T [C \Sigma_{\tilde{x},k}^- C^T + \Sigma_{\tilde{v}}]^{-1}.$$

- Then, anticipating a future step in the development, we compute:

$$\frac{L_k}{\Delta t} = \frac{\Sigma_{\tilde{x},k}^- C^T}{\Delta t} [C \Sigma_{\tilde{x},k}^- C^T + \Sigma_{\tilde{v}}]^{-1} = \Sigma_{\tilde{x},k}^- C^T [C \Sigma_{\tilde{x},k}^- C^T \Delta t + \Sigma_{\tilde{v}} \Delta t]^{-1}.$$

- As $\Delta t \rightarrow 0$, if $\Sigma_{\tilde{x},k}^-$ is finite,

$$\left. \begin{array}{l} C \Sigma_{\tilde{x},k}^- C^T \Delta t \rightarrow 0 \\ \Sigma_{\tilde{v}} \Delta t \rightarrow S_v \\ \Sigma_{\tilde{x},k}^- \rightarrow \Sigma_{\tilde{x}}(t) \end{array} \right\} \text{ or, } \frac{L_k}{\Delta t} \rightarrow L(t) = \Sigma_{\tilde{x}}(t) C^T S_v^{-1},$$

where we recall from Lesson 1.3.7 that $\Sigma_{\tilde{v}} \approx S_v / \Delta t$ as $\Delta t \rightarrow 0$.



Conversion: The estimation-error covariance $\Sigma_{\tilde{x}}(t)$

- We examine the prediction-error covariance time update:

$$\begin{aligned} \Sigma_{\tilde{x},k+1}^- &= A_d \Sigma_{\tilde{x},k}^+ A_d^T + \Sigma_{\tilde{w}} \\ &\approx (I + A \Delta t) \Sigma_{\tilde{x},k}^+ (I + A \Delta t)^T + B_w S_w B_w^T \Delta t \\ &= \Sigma_{\tilde{x},k}^+ + \Delta t \left[A \Sigma_{\tilde{x},k}^+ + \Sigma_{\tilde{x},k}^+ A^T + B_w S_w B_w^T \right] + \mathcal{O}(\Delta t^2). \end{aligned}$$

- In this analysis, we have approximated

$$\begin{aligned} A_d &= e^{A \Delta t} \\ &\approx I + A \Delta t, \end{aligned}$$

like we did in Lesson 1.2.5.

- We also approximated $\Sigma_{\tilde{w}} \approx B_w S_w B_w^T \Delta t$ as we did in Lesson 1.3.6.
- Both of these approximations are valid only when $\Delta t \rightarrow 0$.



Conversion of $\Sigma_{\tilde{x}}(t)$, continued

- Now, we recognize that $\Sigma_{\tilde{x},k}^+ = \Sigma_{\tilde{x},k}^- - L_k C \Sigma_{\tilde{x},k}^-$. Inserting:

$$\begin{aligned} \Sigma_{\tilde{x},k+1}^- &= \Sigma_{\tilde{x},k}^- - L_k C \Sigma_{\tilde{x},k}^- + \Delta t \left[A \Sigma_{\tilde{x},k}^- - A L_k C \Sigma_{\tilde{x},k}^- + \Sigma_{\tilde{x},k}^- A^T \right. \\ &\quad \left. - L_k C \Sigma_{\tilde{x},k}^- A^T + B_w S_w B_w^T \right] + \mathcal{O}(\Delta t^2). \end{aligned}$$

- Rearrange...

$$\begin{aligned} \frac{\Sigma_{\tilde{x},k+1}^- - \Sigma_{\tilde{x},k}^-}{\Delta t} &= \frac{-L_k C \Sigma_{\tilde{x},k}^- + A \Sigma_{\tilde{x},k}^- + \Sigma_{\tilde{x},k}^- A^T + B_w S_w B_w^T}{\Delta t} \\ &\quad - A L_k C \Sigma_{\tilde{x},k}^- - L_k C \Sigma_{\tilde{x},k}^- A^T + \mathcal{O}(\Delta t). \end{aligned}$$

- As $\Delta t \rightarrow 0$, LHS $\rightarrow \dot{\Sigma}_{\tilde{x}}(t)$, $\frac{L_k}{\Delta t} \rightarrow L(t)$, $\Sigma_{\tilde{x},k}^- \rightarrow \Sigma_{\tilde{x}}(t)$ and $A L_k C \Sigma_{\tilde{x},k}^- \rightarrow 0$.



Conversion of $\Sigma_{\tilde{x}}(t)$, final form

- The resulting expression is:

$$\dot{\Sigma}_{\tilde{x}}(t) = \underbrace{A \Sigma_{\tilde{x}}(t) + \Sigma_{\tilde{x}}(t) A^T + B_w S_w B_w^T}_{\text{Lyapunov for propagation}} - \underbrace{\Sigma_{\tilde{x}}(t) C^T S_v^{-1} C \Sigma_{\tilde{x}}(t)}_{\geq 0}.$$

Riccati for measurement update

- This is a continuous-time *differential Riccati equation* for error covariance.
- The equation includes the following impact on $\dot{\Sigma}_{\tilde{x}}(t)$:
 - $A \Sigma_{\tilde{x}}(t) + \Sigma_{\tilde{x}}(t) A^T$ $\Rightarrow$ Homogeneous part.
 - $B_w S_w B_w^T$ $\Rightarrow$ Increase due to process noise.
 - $\Sigma_{\tilde{x}}(t) C^T S_v^{-1} C \Sigma_{\tilde{x}}(t)$ $\Rightarrow$ Decrease due to measurements.



Conversion: State estimate

- Recall the state-prediction and state-estimate steps:

- $\hat{x}_k^- = A_d \hat{x}_{k-1}^+$
- $\hat{x}_k^+ = \hat{x}_k^- + L_k (y_k - C \hat{x}_k^-)$

- Substitute (1) into (2):

$$\hat{x}_k^+ = A_d \hat{x}_{k-1}^+ + L_k (z_k - C A_d \hat{x}_{k-1}^+).$$

- Substitute $A_d \approx I + A \Delta t$:

$$\hat{x}_k^+ = (I + A \Delta t) \hat{x}_{k-1}^+ + L_k (z_k - C(I + A \Delta t) \hat{x}_{k-1}^+).$$

- So, rearranging and taking the limit as $\Delta t \rightarrow 0$:

$$\begin{aligned} \frac{\hat{x}_k^+ - \hat{x}_{k-1}^+}{\Delta t} &= A \hat{x}_{k-1}^+ + \frac{L_k}{\Delta t} (z_k - C \hat{x}_{k-1}^+ - C A \Delta t \hat{x}_{k-1}^+) \\ \dot{\hat{x}}(t) &= A \hat{x}(t) + L(t) [z(t) - C \hat{x}(t)]. \end{aligned}$$



Kalman–Bucy filter equations

- Finally, adding back in the effect of $u(t)$, we have:

$$\dot{\hat{x}}(t) = A\hat{x}(t) + Bu(t) + L(t)[z(t) - C\hat{x}(t) - Du(t)]$$

$$L(t) = \Sigma_{\tilde{x}}(t)C^T S_v^{-1}$$

$$\dot{\Sigma}_{\tilde{x}}(t) = A\Sigma_{\tilde{x}}(t) + \Sigma_{\tilde{x}}(t)A^T + B_w S_w B_w^T - \Sigma_{\tilde{x}}(t)C^T S_v^{-1} C \Sigma_{\tilde{x}}(t).$$

- This is called a “Kalman–Bucy” filter, or a continuous-time Kalman filter/estimator.
- Often very difficult to implement due to the challenges of evaluating the differential Riccati equation in real time.
- But, with a determined attitude, it can be done using operational-amplifier (“analog computer”) circuits.



Steady-state in continuous time

- A steady-state solution is more practical to implement than a time-varying solution for LTI continuous-time systems.
- We let $\dot{\Sigma}_{\tilde{x}}(t) \rightarrow 0$, giving the continuous-time algebraic Riccati equation (ARE):

$$A\Sigma_{\tilde{x}} + \Sigma_{\tilde{x}}A^T + B_w S_w B_w^T - \Sigma_{\tilde{x}}C^T S_v^{-1} C \Sigma_{\tilde{x}} = 0,$$

where we also have $L = \Sigma_{\tilde{x}}C^T S_v^{-1}$.

- Same as time-varying case, but with constant $\Sigma_{\tilde{x}}$ and L . Solve with: `lqe.m`
- The tradeoff between sensor and process noise is now explicit: $L = \Sigma_{\tilde{x}}C^T S_v^{-1}$.
 - If we are uncertain about the estimate (i.e., $\Sigma_{\tilde{x}}$ is large), then the innovation $(z(t) - C\hat{x}(t))$ is weighted heavily (big L).
 - If S_v is small (measurements are accurate) then new measurements are heavily weighted (big L).
- We can think of $\Sigma_{\tilde{x}}C^T S_v^{-1}$ as analogous to a signal-to-noise ratio (SNR).



Error dynamics of the steady-state filter

- We are interested in finding the estimation-error dynamics.
- We start with the plant dynamics:

$$\dot{x} = Ax + B_u u + B_w w$$

$$z = Cx + v$$

where $w \sim \mathcal{N}(0, S_w)$, $B_w S_w B_w^T > 0$, $v \sim \mathcal{N}(0, S_v)$, $S_v > 0$; w, v white and independent, $[A, C]$ observable. (Strong form of assumptions.)

- Steady-state Kalman filter: $\dot{\hat{x}} = A\hat{x} + B_u u + L(z - C\hat{x})$.
- This gives estimation error dynamics: $\tilde{x} = x - \hat{x}$

$$\begin{aligned} \dot{\tilde{x}} &= \dot{x} - \dot{\hat{x}} = Ax + B_w w - [(A - LC)\hat{x} + L(Cx + v)] \\ &= (A - LC)\tilde{x} + B_w w - Lv. \end{aligned}$$

- Filter stability governed by eigenvalues of $(A - LC)$.



Frequency-domain interpretation

- This equation makes explicit a tradeoff between:
 - Speed of error decay (L big, $\text{eig}(A - LC)$ far in LHP).
 - Susceptibility of error to corruption by sensor noise (L small so Lv small).
- Kalman filter selects the *optimal balance* between these two goals.
- Can view in the frequency domain: Have $\hat{\hat{x}} = A\hat{x} + L(z - C\hat{x})$
- Laplace transform both sides... (assume scalar state for simplicity):

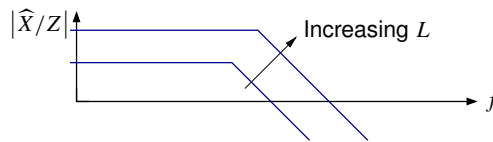
$$\frac{\hat{X}(s)}{Z(s)} = \frac{L}{s - A + LC} = \frac{\frac{L}{LC - A}}{\frac{s}{LC - A} + 1}.$$

- Pole at $s = -(LC - A)$ and dc-gain of $L/(LC - A)$.
- This is the transfer function of the filter applied to the measurements to form the estimate $\hat{x}$ (a low-pass filter!).



Visualizing filter frequency response

- Increasing L ($\Sigma_{\tilde{x}}$ is large or S_v is small) pushes the filter magnitude response up and out.



- Eventually, the estimate would be too corrupted by the noise in the measurements; this is why the Kalman filter must choose an optimal L .
- Note that balancing the sensor-noise impact is done with respect to the process noise ($B_w w$), which is implicitly present in $\Sigma_{\tilde{x}}$.
- It turns out that the ratio S_w/S_v plays a key role in the selection of L , as an example will demonstrate.



Example of a time-varying filter

- EXAMPLE: System driven by noise:

$$\begin{aligned}\dot{\hat{x}}(t) &= w(t) \\ z(t) &= x(t) + v(t),\end{aligned}$$

where $\mathbb{E}[w] = \mathbb{E}[v] = 0$ and w and v independent; $\mathbb{E}[w(t)w(t + \tau)] = S_w\delta(\tau)$ and $\mathbb{E}[v(t)v(t + \tau)] = S_v\delta(\tau)$. Also, $\mathbb{E}[x(0)] = 0$, $\mathbb{E}[x(0)^2] = \Sigma_{\tilde{x}}(0)$.

- The optimal time-varying filter is:

$$\dot{\hat{x}}(t) = \frac{\Sigma_{\tilde{x}}(t)}{S_v}(z(t) - \hat{x}(t)), \quad \text{and} \quad \Sigma_{\tilde{x}}(t) = \sqrt{S_w S_v} \frac{1 + b e^{-2\alpha t}}{1 - b e^{-2\alpha t}},$$

where $\alpha = \sqrt{S_w/S_v}$ and $b = (\Sigma_{\tilde{x}}(0) - \sqrt{S_w S_v})/(\Sigma_{\tilde{x}}(0) + \sqrt{S_w S_v})$.

- Convergence speed of $\Sigma_{\tilde{x}}(t)$ is determined by S_w/S_v via α .



Example of steady-state filter

- In steady-state, the Kalman-filter equations become:

$$\Sigma_{\tilde{x}}(t) \rightarrow \Sigma_{\tilde{x},ss} = \sqrt{S_w S_v}$$

$$L(t) \rightarrow L_{ss} = \frac{\Sigma_{\tilde{x},ss}}{S_v} = \sqrt{S_w/S_v}.$$

- Therefore, the steady-state filter implements:

$$\dot{\hat{x}}(t) = -\sqrt{\frac{S_w}{S_v}} \hat{x}(t) + \sqrt{\frac{S_w}{S_v}} z(t),$$

and the closed-loop pole location is determined by S_w/S_v .

- If S_w/S_v small, sensors are relatively noisy, state converges slowly.
- If S_w/S_v large, sensors relatively clean, state converges quickly.



Summary

- You learned that we can develop Kalman-filter equations in continuous-time, resulting in the Kalman–Bucy filter.
- It is generally difficult to perform these calculations (e.g., solving the continuous-time differential Riccati equation) in real time although it can be done using operational-amplifier (“analog computer”) circuits.
- More often, a steady-state Kalman–Bucy filter is implemented, where the Kalman gain is computed as the solution to a continuous-time algebraic Riccati equation.
 - Since this gain is constant, it is easier to implement this solution using analog circuits.
- An example showed a frequency-domain interpretation of the Kalman–Bucy filter, and how it acts as an optimal low-pass filter on noisy measurements.
- The example also demonstrated the impact of the ratio S_w/S_v on the convergence speed of the filter and the product $S_w S_v$ on the confidence of the final answer.